

Artificial Intelligence Cloud Innovation Center Discovery Workshop Readout



W. P. Carey School of Business

W. P. Carey Center for Real Estate and Finance



Date of Discovery Workshop: 6/15/26

powered by **aws**


1. Introduction

The W. P. Carey School of Business Real Estate Program at Arizona State University is exploring the development of an AI-enabled experiential learning platform designed to teach real estate development processes through interactive project simulation and structured decision-making exercises.

This initiative builds upon an existing instructional methodology developed within the Real Estate Program that teaches students how complex real estate projects evolve across multiple phases and stakeholder groups. The framework centers around four foundational dimensions: **Product, People, Purpose, and Process**, helping students understand how development decisions, participants, constraints, and project goals interact throughout the lifecycle of a real estate project.

Historically, this experience has been delivered through in-person exercises using physical cards, process maps, and collaborative scenario-building activities where students organize project components, assign responsibilities, build schedules, and navigate changing project conditions.

This project seeks to transform that instructional model into an AI-supported digital platform that enables repeatable learning, scalable classroom delivery, automated feedback, and more immersive experiential education aligned to the goals of the W. P. Carey Real Estate Program.

PROBLEM

Real estate development education requires students to understand highly interconnected systems involving timelines, stakeholders, regulations, budgets, negotiations, sequencing, and evolving business conditions.

Traditional teaching methods create several limitations:

- Difficulty simulating real-world complexity across multiple development phases
- Limited scalability of physical workshop exercises
- Challenges measuring student decision-making and process understanding
- High instructor effort to facilitate and assess experiential activities
- Limited repeatability across cohorts and scenarios

Although the underlying real estate development process remains structured, each project contains unique variables—including municipal requirements, property conditions, financing assumptions, stakeholder priorities, and timing considerations.

The program seeks to create a digital learning environment that allows students to repeatedly practice real estate development workflows while receiving structured guidance and measurable feedback.

2. POC Scope

2.1 Objectives:

- Digitize the existing real estate development instructional framework into an interactive educational platform
- Enable students to model and navigate real estate development processes using Product, People, Purpose, and Process principles
- Create repeatable project-based learning experiences for real estate students
- Support instructor-driven scenario creation and classroom delivery
- Provide automated evaluation and feedback mechanisms
- Establish scalable infrastructure capable of supporting broader experiential learning initiatives across the W. P. Carey School of Business

2.2 Scope:

The following items shall be **IN SCOPE** of this project:

- Interactive student workspace supporting drag-and-drop real estate development exercises
- Instructor authoring capabilities including:
 - Scenario creation
 - Development phase configuration
 - Activity and dependency management
 - Stakeholder definition
 - Evaluation settings
- Student workflows including:
 - Scenario participation
 - Draft saving
 - Submission and revision
 - Progress tracking
 - Guided reflection
- AI-supported functionality including:
 - Requirement validation
 - Scenario guidance
 - Learning feedback
 - Progress analysis

- Configuration verification
- Real estate project simulation features including:
 - Multi-phase project organization
 - Activity placement across phases
 - Stakeholder and role mapping
 - Budget and schedule representation
 - Process visualization
- Classroom functionality including:
 - Individual and team participation
 - Live classroom support
 - Historical progress tracking
 - Assessment support
- Serverless cloud deployment architecture
- Instructor and student web applications

2.3 Out of Scope:

The following items shall **NOT** be under CIC development plan:

- Canvas gradebook integration (future enhancement)
- Enterprise SSO implementation during initial deployment
- Commercial project management software replacement
- General-purpose PM certification training
- Native mobile application development
- Full institutional LMS integrations
- Licensing and commercialization models

2.4 Success Metrics:

- Students successfully complete project simulations independently
- Reduction in instructor effort for setup and evaluation
- Positive usability feedback from students and instructors
- Ability to support at least 100 concurrent users
- Successful execution of repeatable exercises across multiple scenarios
- Improved student understanding of project lifecycle concepts
- Stable deployment and operation within AWS infrastructure
- Successful generation of reusable instructional templates

2.5 Assumptions:

- Existing instructional methodology remains foundational to the digital experience
- Students benefit from experiential and visual learning approaches
- Instructors will provide content, branding, and exercise configuration guidance
- AI-generated recommendations will augment—not replace—instruction
- Serverless architecture provides sufficient scalability and cost flexibility
- Future institutional integrations may occur after initial validation

2.6 Deliverables:

- Functional educational web application
- Student and instructor portals
- Interactive card-based project exercise environment
- AI-assisted requirement verification workflows
- Submission and scoring engine

- Results and history tracking
- Technical architecture documentation
- Deployment guidance and operational documentation
- Branding implementation

2.7 Dependencies:

- Instructor-provided educational materials and examples
- Branding guidelines and design assets
- AWS account provisioning
- Availability of deployment infrastructure
- User testing participation
- Final approval of generated requirements and user stories

2.8 Risk:

- Scope growth from expanding instructional requirements
- Complexity of translating physical learning exercises into digital workflows
- Adoption challenges among instructors and students
- AI context limitations impacting generation quality
- User experience complexity from highly configurable workflows
- Performance constraints during classroom-scale usage

2.9 Data:

The following datasets and materials are expected to support the POC:

Sl.no	Dataset	Format	Source	Notes
1	Process framework materials	documents	instructor	Educational structure
2	Exercise templates	structured	instructor	Assignment generation
3	Student submissions	Application data	students	Scoring and tracking
4	User activity data	System generated	platform	Progress analysis

2.10 Challenges

- Preserving instructional intent while digitizing physical experiences
- Designing intuitive project visualization experiences
- Supporting flexible project structures and repeated activities
- Balancing customization with usability
- Maintaining responsiveness during collaborative sessions
- Designing meaningful automated feedback mechanisms

2.11 Timeline

The following is a proposed high-level outline of key tasks that the ASU/AWS team will undertake in close collaboration and coordination with the customer team. Please note, the plan is subject to change as additional clarity on data and other factors emerge. Engagement updates and progress will be routinely delivered and shared during regular meetings between customer, AWS, and ASU teams.

Item #	Iterations	Deliverables	Date
1	Inception	Requirements validation	done
2	Design	Application design and UI generation	TBD
3	Build	Core platform development	TBD
4	Deploy	AWS deployment and testing	TBD

2.12 Future Scope

- Canvas integration and gradebook synchronization
- Enterprise SSO and institutional identity support
- Expanded classroom collaboration capabilities
- Additional assessment analytics and dashboards
- AI-powered exercise generation
- Marketplace of reusable project templates
- Multi-institution educational deployments
- Expanded deployment to underserved educational communities

3 POC Execution

3.1 Sprint Cadence

The ASU AI CIC powered by AWS delivery methodology is primarily Scrum-based. The executing team works on a weekly or bi-weekly cadence. Normally there is at least a weekly or bi-weekly touchpoint with the customer team and stakeholders for project updates and feedback.

3.2 Customer Owners

This section specifies the key owners and points of contact on customer team:

- Mark Stapp
- Tricia Dunlavey Hilbun

3.3 Next Steps

To continue progress on the POC engagement, the following general plan is proposed which will be continually updated and communicated in close collaboration with customer stakeholders.

1. Establish a cadence schedule between WP Carey stakeholders and ASU/AWS teams to review and share POC progress.
2. WP Carey team to provide relevant datasets to ASU team.
3. Upon a successful POC, secure approval(s) for public reference from customer executive sponsor and co-creation of reference and/or case study materials related to the POC, including but not limited to
 - a. Quote
 - b. [ASU.edu](https://www.asu.edu) blog post
 - c. AWS Event Co-Presentation, logo use and use case mentions

4. Additional Notes

The following additional notes and assumptions were noted during the Discovery Workshop and follow-up session and will continue to be updated as POC development progresses:

- This is a living document and is subject to change. All changes will be discussed and agreed upon during regular weekly meetings between ASU AI CIC and WP Carey teams.